



AN EMULATOR SOLUTION HELPED DCC TO IMPLEMENT SMART METERING SYSTEMS ON TIME

System Level Solutions had provided the emulator solution for integration testing during to execute Smart Metering Implementation Programme

Industry

Data Communication & Networking

Challenges

- Great pressure to execute Smart Metering Implementation program on time and deliver fully functional CH
- No possibility to identify CH behaviour in different real-world situations due to lack of full-fledged Communications Hub testing in Smart meters

Solution

Emulator Solution to execute Smart Metering Implementation Programme.

Features

- Emulates ESME, GSME, HCALCS, PPMID and IHD HAN Device type
- Supports GBCS2.0 and SMETS3 Technical Specifications.
- Supports communication protocols like Zigbee, DLMS and ASN
- Supports retaining configuration state on certain transactions without power reset
- Provides the capability to simulate noisy HAN environments or coexist with other RF signal generation equipment which can be used to inject noise
- Configurable fields to enable full coverage of GBCS message structures

Outcome

- Integration testing during smart meter implementation
- Identification of CH behaviour in real environment

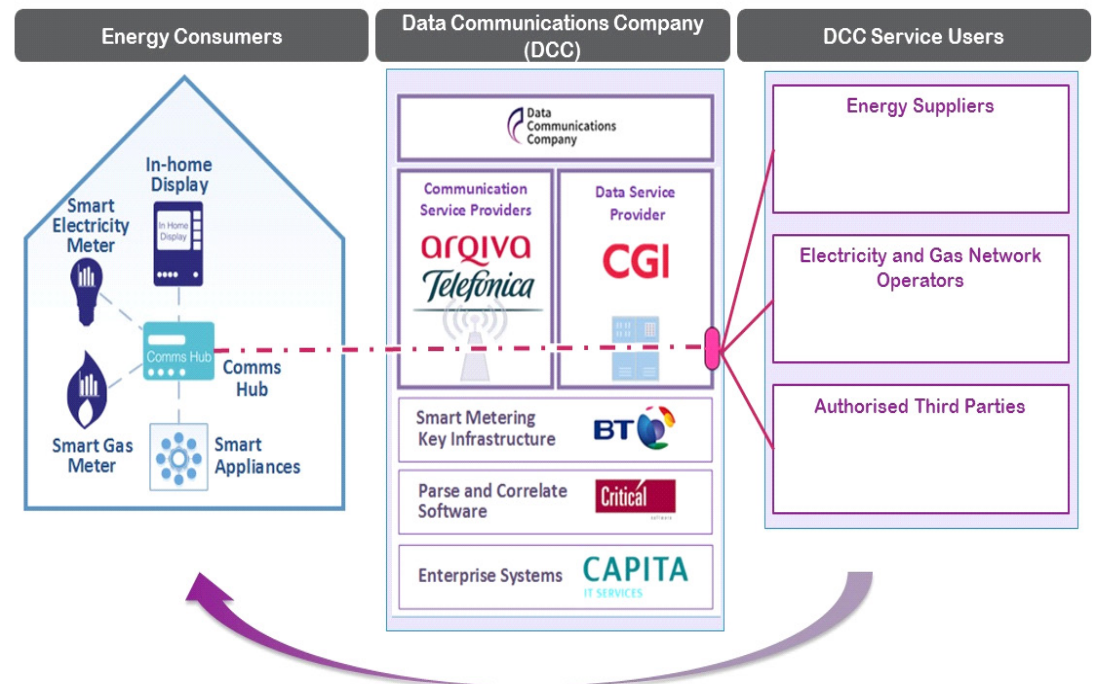


Figure 1: An overview of DCC operational infrastructure

Summary

SLS made Smart Meter Emulator which operates in Dual Frequency Bands i.e, 2.4GHz band and Sub-GHz band where 2.4GHz was for ESME related functions while GSME and all other devices are for both bands i.e 2.4GHz and Sub-GHz

Data Communication Company (DCC) or Smart DCC was looking for an Emulator Solution to execute Smart Metering Implementation Programme. SLS made Smart Meter Emulator which operates in Dual Frequency Bands i.e, 2.4GHz band and Sub-GHz band where 2.4GHz was for ESME related functions while GSME and all other devices are for both bands i.e 2.4GHz and Sub-GHz. The emulator is one of the key devices in DCC's Integration Testing Program that helped to achieve the major milestone to execute Integration Testing PIT and SIT phase of R2.0 DBCH Testing.

Client Profile

Data Communication Company (DCC): Smart DCC has built and maintained the secure national infrastructure that underpins the roll-out of smart meters across Great Britain. This wireless network connects smart meters to energy suppliers, network operators and other authorised service users. It is maintained to very high security standards, as endorsed by the National Cyber Security Centre. The data communication network infrastructure (Ref: Figure 1) is being delivered on the ground by two Communications Services Providers, Arqiva and Telefónica, and a Data Services Provider, CGI.

Business Challenge

DCC delivers services through procuring and contracting with Service Providers and assuring delivery of those services, during both implementation and live operations. As part of this strategy, DCC awarded the contract to Arquiva and TEF to provide the communications infrastructure that connects smart meters to DCC systems. This comprises the nationwide Smart Metering Wide Area Network (SM WAN) and the Communications Hubs that establish the Home Area Network in the consumer's premises.

Communications Hubs needs to be compliant with various standards like GBCS, SMETS, etc. and should support protocols like Zigbee, ASN and DLMS to communicate with other HAN devices connected with it. So to examine all these implementations correctly and to verify interoperability between CH (Communications Hub) and various Smart Meter Manufacturers, DCC implemented Integration Testing program for CH and Meter Manufacturers.

DCC proposed Integration Testing program with various stages (PIT, SIT and UIT) in which CH and various Smart Meter Manufacturers test their device in the Real environment. However, DCC has faced following challenges to execute this plan.

- Compare to CH Manufacturers, Meter Manufacturers were lagging behind in implementation of various standards like GBCS, SMETS, etc. Hence, DCC observed great pressure to execute Smart Metering Implementation program on time and deliver fully functional CH.

- Available Smart Meters also did not support full-fledged Communications Hub testing including execution of various negative test cases to identify CH behavior in different real-world situations.

The Solution

DCC issued their requirements in the form of CR (Change Request) and SLS then completed full impact assessment which includes cost, time, resources and other critical information. DCC reviewed impact assessment submitted by SLS and awarded contract based on assessment. Similarly, Telefonica and Arquiva had also awarded contract to SLS for manufacturing Emulators according to requirements of DCC.

The deciding factors in favour of SLS were,

- Cost and time to deliver Solution
- Expertise in Smart Metering Domain
- Emulator Solution provided flexibility to include new Change requests in existing solution quickly and efficiently without altering existing functionality.

SLS made Smart Meter Emulator which operates in Dual Frequency Bands i.e, 2.4G band Sub-GHz band where 2.4G was for ESME related functions and Sub-GHz for GSME related functions. It emulates various devices used in smart metering system such as Gas Smart Meter Equipment (GSME), Energy Smart Meter Equipment (ESME), Pre-Payment Meter Interface Device (PPMID), In-Home Display (IHD) and Auxiliary Load Control Switch (ALCS). It can send/receive the ZigBee commands in HAN network, acting as ESME, GSME or any of above device types. It contains the LCD and CapSense buttons to control and indicate the details while emulating. SLS Emulator Solution has been used by DCC to execute Smart Metering Implementation Programme.

Emulator Solutions emulates all five HAN Devices within Single Hardware which reduce significant cost and space compare to buying five different HAN Devices. It came with configurable User Interface to load and verify test data with minimum complexity. Moreover,





Emulator Solution provided flexibility to include new Change requests in existing solution quickly and efficiently without altering existing functionality. SLS was solely responsible for the solution and has executed all the Hardware and Software Life Cycles (Design, Development, Testing and Support) exceptionally well.

SLS provides efficient onsite/offsite support with the lowest turn-around time among all the contracted CSPs of DCC

Technical Features

As the GBCS specification gets changes at interval, there is a requirement to comply all the devices to new specification before they installed. Emulator Solution allows CH manufacturers to demonstrate the full testing of GBCS specification releases, without the need to test on real smart meters and other HAN devices, functionally on required band i.e. 2.4 GHz and 868 Mhz.

Technical details are given below:

- Emulates ESME, GSME, HCALCS, PPMID and IHD HAN Device type
- 2.Supports GBCS2.0 and SMETS3 Technical Specifications.
- Supports communication protocols like Zigbee, DLMS and ASN
- Supports retaining configuration state on certain transactions without power reset
- Provides the capability to simulate noisy HAN environments or coexist with other RF signal generation equipment which can be used to inject noise
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Emulator helped DCC to achieve the major milestone to execute Integration Testing PIT and SIT phase of R2.0 DBCH Testing.

Outcome

SLS fulfilled all the expectations of DCC. In fact, SLS is the only DCC contracted CSP who has 100% record in terms of timely delivery. This in turn helped DCC to fulfil its milestone. DCC rated SLS performance as "Satisfactory" in its Annual Service Report. With the help of Emulator Solution, DCC completed the major milestone of PIT and SIT completion of R2.0 DBCH Testing on time. Now, DCC is on track to deliver R2.0 DBCH Solution as per their forecast and also impressed their investors with this performance.

SLS Emulator Solution has been used by DCC to execute Smart Metering Implementation Programme. Emulator is one of the key devices in DCC's Integration Testing Program. It helped DCC to execute Smart Metering Implementation program on time as DCC has right to govern the software solution in terms of functionality, delivery and testing. This solution can also be configurable for various negative test cases which help DCC to identify CH behaviour in real environment.

"SLS is the first organization to perceive Zigbee certifications for Smart Metering Devices under Smart Energy category."



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